

04 Cheat Sheets

Complexity Cheat Sheet

Pattern	Time	Space	Notes
HashMap insert/lookup/delete	$O(1)$	$O(U)$	U = unique entries
Set insert/lookup	$O(1)$	$O(U)$	
Heap push	$O(\log K)$	$O(K)$	
Heap top-K total	$O(N \log K)$	$O(K)$	Keep heap size = K
Binary Search	$O(\log N)$	$O(1)$	Requires sorted data
Sliding Window	$O(N)$	$O(K)$	K = window state
DFS (time)	$O(N)$	—	Every node visited once
DFS (space, balanced)	—	$O(\log N)$	Recursion stack
DFS (space, skewed)	—	$O(N)$	Recursion stack
BFS	$O(V + E)$	$O(V)$	
Two Pointers	$O(N)$	$O(1)$	
Sort	$O(N \log N)$	$O(N)$	
Intervals (merge)	$O(N \log N)$	$O(N)$	Dominated by sort
Cosine Similarity	$O(D)$	$O(1)$	D = embedding dim
Top-K Retrieval	$O(ND + N \log K)$	$O(K)$	N = docs, D = dim
Chunking	$O(N)$	$O(N)$	
Context Truncation	$O(N)$	$O(N)$	
Near-Duplicate (pairwise)	$O(N^2D)$	$O(1)$	All pairs comparison

ML Coding Interview Checklist

Can you implement from memory?

DSA Patterns

- ☐ HashMap (frequency count, duplicate detection)
- ☐ Heap (top-K, kth largest)
- ☐ Sliding Window (fixed + variable)
- ☐ Binary Search (standard + lower/upper bound)
- ☐ Two Pointers (opposite direction)
- ☐ DFS on trees (max depth, count, sum, search)
- ☐ BFS level order
- ☐ Merge intervals

ML/LLM Coding

- ☐ Cosine similarity
- ☐ Dot product
- ☐ Euclidean distance
- ☐ Top-K retrieval with heap
- ☐ Text chunking (fixed + sliding window)
- ☐ Context window truncation
- ☐ Deduplication (exact + near-duplicate)
- ☐ Precision / Recall / F1
- ☐ Recall@K
- ☐ MRR
- ☐ Reranking

Before writing code

1. What pattern fits?
2. What data structure?
3. What are the edge cases?
4. What's the complexity?

Pattern Keywords — Quick Lookup

Keywords	Pattern
frequency, count, duplicate	HashMap
unique, seen before	Set
top k, ranking, priority	Heap
substring, subarray, longest, shortest	Sliding Window
sorted array, threshold, first/last	Binary Search
overlap, meetings, schedule	Intervals
tree, depth, path, hierarchy	DFS
level order, shortest path, BFS	BFS
pair in sorted array, two sum	Two Pointers
embeddings, similarity, semantic search	Cosine Similarity + Heap
context window, token limit	Greedy Reverse
chunk, split documents	Sliding Window
deduplication	Set (exact) / Similarity (near)
ranking quality	Recall@K, MRR, NDCG
RAG, retrieval, knowledge base	Embed → Similarity → Top K → Rerank